The LNM Institute of Information Technology, Jaipur
Software Development Laboratory
Course Code: CS314
Lab Assignment 1: Solution

1. Project Identification

Item	Details
Project Name	Hostel Complaint Management System
Project Type	Web-based Hostel Complaint Management and Tracking System
Team ID	T042
Team Members	DHRUV AGRAWAL, SANYAM HUMAD, HARSHIL AGRAWAL, AKSHAT KHANDAL
Roll Numbers	24UCS016, 24UCS017, 24UCS018, 24UCS019
Development Model	Agile + Incremental Development
Primary Users	Students, Hostel/Warden Authorities, Maintenance Staff, Administrators
Development Team	Student software development team
Course	Software Development Lab
Project Objective	To develop a centralized platform for reporting, managing, tracking and resolving hostel complaints
Expected Outcome	A working, tested and documented software prototype developed incrementally

2. Project Background Current Problem

Current Problem

Hostel-related complaints is communicated through disconnected or informal channels such as verbal communication, messages, calls, forms or manual records.

Such a process can make it difficult to:

- maintain complaint history,
- identify pending complaints,
- assign responsibility,
- track progress,
- communicate complaint status,

- monitor unresolved issues,
- generate useful summaries

Proposed Solution

The Hostel Complaint Management System will provide a centralized platform where complaints can be: submitted - recorded - categorized - reviewed - assigned - worked on - updated - resolved/closed. Students will be able to track their complaints, while authorized hostel personnel will be able to manage complaints and monitor their status.

3. Project Vision

The Hostel Complaint Management System aims to provide a centralized, transparent, userfriendly and reliable platform for reporting, tracking, managing and resolving hostel complaints, improving visibility of complaint status for students and enabling authorized hostel personnel to manage maintenance requests systematically.

4. Problem Statement

Hostel residents may face maintenance and facility-related problems that need to be communicated to the appropriate hostel authority or maintenance personnel. When complaint handling is distributed across informal or disconnected communication channels, complaints can be difficult to track, prioritize, assign and close systematically. Students may also lack a clear way to know whether a complaint has been received, is being worked on, or has been resolved. Hostel authorities may lack a centralized record for monitoring pending and resolved complaints and generating useful summaries. The proposed system addresses this problem by providing an integrated platform for complaint submission, management, tracking, status updates and reporting.

5. Project Objectives

The initial objectives should be:

- To provide a centralized platform for students to submit hostel-related complaints.
- To allow complaints to be categorized using suitable complaint types or categories.
- To allow authorized hostel personnel to view and manage submitted complaints.
- To support assignment of complaints to appropriate maintenance personnel where applicable.
- To provide complaint status tracking so that progress can be monitored.
- To maintain complaint records and history for future reference and monitoring.

- To provide relevant notifications or status updates wherever included in the implemented scope.
- To provide basic reports or dashboards for authorized users to monitor complaint activity.
- To enforce appropriate role-based access to system functionality.
- To develop, test, document and release the prototype incrementally using sound software engineering practices.

6. Stakeholders

Stakeholder	Interest / Role
Students	Submit complaints, provide details, track status and view complaint history/outcomes.
Hostel/Warden Authority	Review, manage, prioritize and monitor complaints; assign responsibility where applicable.
Maintenance Staff	Handle assigned complaints and update progress/resolution information.
Administrator	Manage users, roles and oversee overall system
Institute/Management	Monitor complaint activity and use authorized reports/summaries.
System Development Team	Analyse, design, implement, test, document and maintain the prototype.

7. Initial Scope In Scope :

- User registration/login
- Role-based access
- Complaint submission
- Complaint categorization
- Complaint viewing and management
- Complaint assignment
- Complaint status tracking
- Complaint history
- Notifications/status updates
- Basic reports/dashboard

Initially Out of Scope :

- Payment processing
- Hostel admission management
- Complete room-allocation management
- Mess/food management
- Complete college ERP
- Automated physical repair
- Hardware/IoT control
- Advanced AI-based diagnosis/recommendation
- Academic course registration
- Online examinations

8. Initial Assumptions

- Users have access to the system through an available network/internet connection.
- Users have valid credentials/accounts appropriate to their roles.
- Students are authorized to submit complaints concerning hostel facilities/services.
- Hostel authorities are authorized to view and manage complaints within their responsibility.
- Maintenance personnel can receive or view complaints assigned to them.
- Complaint status is updated by authorized users.
- The initial system will be developed as a web application.
- The system will use a centralized database for user and complaint records.

9. Initial Constraints

- Limited semester/development time.
- Four-member student development team.
- Limited access to real hostel/institutional complaint data.
- Limited testing with large numbers of real hostel users.
- Limited development infrastructure/resources.
- Requirements may change during development.
- User and complaint data must be handled with appropriate privacy/security considerations.
- The final prototype must remain manageable enough to demonstrate and test within the course timeline.

10. Initial Success Criteria

- Different user roles can access appropriate functionality.
- A student can authenticate and submit a hostel complaint.
- The complaint is stored correctly.
- An authorized hostel user can view and manage the complaint.

- A complaint can be assigned to responsible personnel where applicable.
- The complaint can move through its defined status workflow.
- A student can view the current status/outcome.
- Complaint history is maintained correctly.
- Basic authorized reports/dashboard information can be demonstrated if included in scope.
- Important system functions are tested systematically.
- Defects are documented and addressed.
- The project is maintained in the designated Git/GitHub repository with meaningful development history.
- The final prototype has appropriate software-engineering documentation.

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Lab Assignment 2: Solution

11. User Classes

11.1 Student

Students are the primary participants who report hostel-related problems.
They shall be able to:

- authenticate
- submit complaints
- provide complaint details
- select complaint categories
- view their submitted complaints;
- track complaint status;
- view complaint history and outcomes
- receive/view relevant status updates
- view resolution information

11.2 Hostel/Warden Authority

The Hostel/Warden Authority is responsible for reviewing and managing hostel complaints.

- review submitted complaints;
- view complaint details;
- categorize and prioritize complaints wherever applicable;
- assign complaints to maintenance personnel;
- monitor complaint progress;
- manage complaint status where authorized;
- close resolved complaints;
- view complaint history;
- access authorized complaint reports

11.3 Maintenance Staff

Maintenance Staff are responsible for handling complaints assigned to them.

- view assigned complaints;
- view complaint details;
- update complaint progress;
- record work/resolution information;
- mark work as completed;
- view relevant complaint history.

11.4 Administrator

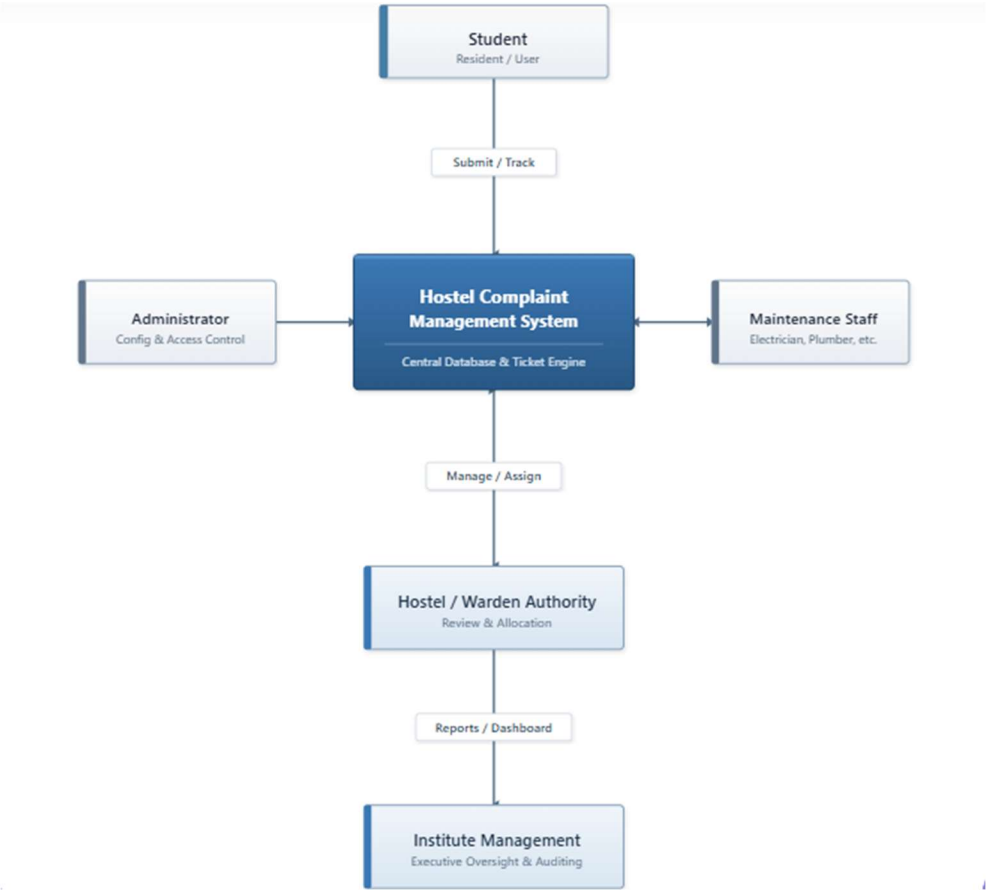
The administrator shall:

- manage users;
- assign and manage roles;
- manage account status;
- monitor system activity;
- access administrative reports.

11.5 Institute Management

Institute Management users shall have access to appropriate aggregated reports and dashboards for monitoring hostel complaint activity.

12. System Context



13. Functional Requirements

13.1 Authentication and Account Management

FR-001 -User Login
The system shall allow a registered user to authenticate using valid credentials.

FR-002 -Authentication Failure
The system shall notify the user when authentication fails without unnecessarily revealing sensitive authentication information.

FR-003 -Logout
The system shall allow authenticated users to terminate their session.

FR-004 -Role Assignment
The system shall allow authorized administrators to assign roles to users.

FR-005 -Role-Based Access
The system shall restrict system functions according to the user's assigned role.

13.2 Complaint Submission

FR-006 -Submit Complaint

The system shall allow an authenticated student to submit a hostel complaint.

FR-007 -Complaint Information

The system shall allow a student to provide:

- complaint category;
- complaint title;
- complaint description;
- hostel/block information where applicable;
- room/location information where applicable;

System records:

- Submission Date/Time

FR-008 -Complaint Status
The system shall assign the initial status Submitted to a successfully created complaint.

FR-009 -Complaint ID
The system shall generate a unique identifier for each successfully submitted complaint.

FR-010 -Complaint Validation
The system shall validate mandatory complaint information before creating a complaint record.

13.3 Complaint Review and Management

FR-011 -Review Complaint

The system shall allow an authorized Hostel/Warden Authority to view and review submitted complaints.

FR-012 -Complaint Categorization
The system shall allow an authorized user to categorize a complaint using available complaint categories.

FR-013 -Assign Complaint
The system shall allow an authorized Hostel/Warden Authority to assign an eligible complaint to appropriate maintenance staff.

FR-014 -View Assigned Complaints
The system shall allow maintenance staff to view complaints assigned to them.

13.4 Complaint Progress and Resolution

FR-015 -Update Complaint Progress
The system shall allow authorized maintenance staff to update the progress of an assigned complaint.

FR-016 -Update Complaint Status
The system shall allow authorized users to update complaint status according to the defined complaint lifecycle.

FR-017 -Record Resolution
The system shall allow authorized maintenance staff or hostel authorities to record resolution details for a complaint.

FR-018 -Close Complaint
The system shall allow an authorized Hostel/Warden Authority to close a complaint after resolution information has been recorded.

13.5 History and Notification

FR-019 -Complaint History
The system shall maintain a history of significant complaint activities and status changes.

FR-020 -Status Notification
The system shall provide relevant complaint status updates to the concerned user.

13.6 Report

FR-021 -Complaint Reports
The system shall provide authorized users with complaint reports containing relevant complaint information and summary statistics.

14. Business Rules

ID	Business Rule
BR-001	Only authenticated users can access protected system functions.
BR-002	Only students can submit complaints for themselves.
BR-003	Every complaint shall have a unique complaint ID.

BR-004	A complaint must contain all mandatory information before submission.
BR-005	Only authorized Hostel/Warden Authority users can assign complaints.
BR-006	Maintenance staff can update progress only for complaints assigned to them.
BR-007	A complaint must be assigned before maintenance staff can update its progress.
BR-008	A complaint cannot be closed without recorded resolution information.
BR-009	Closed complaints cannot be modified through the normal complaint workflow.
BR-010	Only authorized users can access complaint reports.
BR-011	Significant complaint status changes shall be recorded in complaint history.
BR-012	Users shall only be able to access complaint information permitted by their role.

15. Non-Functional Requirements

15.1 Performance

NFR-001: The system should provide acceptable response times for normal user operations under the expected workload.

NFR-002: Complaint search and filtering operations should return results within an acceptable time under expected workload.

NFR-003: The system should support concurrent users appropriate to expected hostel usage.

15.2 Security

NFR-004: Protected system functionality shall require authentication.

NFR-005: Access to system functionality shall be controlled using user roles.

NFR-006: Sensitive user and complaint information shall not be exposed to unauthorized users.

NFR-007: Administrative and significant complaint-management actions should be auditable.

15.3 Usability

NFR-008: The system shall provide consistent navigation across major user interfaces.

NFR-009: Error messages shall be understandable to users.

NFR-010: Common activities such as submitting and tracking complaints should require minimal unnecessary interaction.

15.4 Reliability

NFR-011: Complaint records shall be maintained reliably during normal system operation.

NFR-012: Complaint history records shall not be lost during normal system operation.

15.5 Availability

NFR-013: The system should be available during normal hostel operating periods except during planned maintenance.

15.6 Maintainability

NFR-014: The software shall follow a modular architecture.

NFR-015: Source code shall be maintained using version control.

NFR-016: Software documentation shall be maintained alongside relevant system changes.

15.7 Scalability

NFR-017: The architecture should allow the system to support increasing numbers of users and complaints.

15.8 Compatibility

NFR-018: The system should operate correctly on commonly used modern web browsers.

15.9 Privacy

NFR-019: Personal information shall be accessible only to authorized users.

15.10 Auditability

NFR-020: Significant administrative actions should be recorded to support accountability and troubleshooting.

16. Data Requirements

The initial logical information requirements include the following entities.

User
Possible attributes:

- User ID
- Name
- Email
- Contact information
- Role
- Account status

Complaint
Possible attributes:

- Complaint ID
- Student/User ID
- Category
- Title
- Description
- Hostel/Block
- Room/Location
- Priority
- Submission Date/Time
- Current Status

Complaint Assignment
Possible attributes:

- Assignment ID
- Complaint ID
- Staff ID
- Assigned By
- Assignment Date/Time

Complaint Status History
Possible attributes:

- History ID
- Complaint ID
- Previous Status
- New Status
- Changed By
- Change Date/Time
- Comments

Resolution
Possible attributes:

- Resolution ID
- Complaint ID
- Resolution Details
- Resolved By
- Resolution Date/Time

Notification
Possible attributes:

- Notification ID
- Recipient ID
- Complaint ID
- Message
- Notification Type
- Read Status
- Date/Time

Complaint Category
Possible attributes:

- Category ID
- Category Name
- Description
- Active Status

17. Complaint Lifecycle
The event entity may pass through the following states:

```
Submitted
   ↓
Under Review
   ↓
Assigned
   ↓
In Progress
   ↓
Resolved
   ↓
 Closed
```

18. Major Use Cases

The initial use-case catalogue is as follows:

ID	Use Case	Primary Actor
UC-001	Login	All users
UC-002	Manage Users	Administrator
UC-003	Submit Complaint	Student
UC-004	View Complaint Status	Student
UC-005	Review Complaint	Hostel/Warden Authority
UC-006	Assign Complaint	Hostel/Warden Authority
UC-007	Update Complaint Progress	Maintenance Staff
UC-008	Resolve Complaint	Maintenance Staff / Authority
UC-009	Close Complaint	Hostel/Warden Authority
UC-010	View Complaint History	Student / Authority
UC-011	View Notification	Student / Staff / Authority
UC-012	Generate Complaint Reports	Authorized User

19. Detailed Use Case Example

UC-003 -Submit Complaint

Primary Actor: Student

Preconditions

- Student is authenticated.
- Complaint submission functionality is available.

Main Flow

1. Student selects Submit Complaint.
2. System displays the complaint form.
3. Student enters complaint information.
4. Student selects a complaint category.
5. Student submits the complaint.
6. System validates mandatory information.
7. System generates a unique complaint ID.
8. System creates the complaint record.
9. System sets the status to Submitted.
10. System displays a submission confirmation.

Alternative/Exception Flows

- If mandatory information is missing, the system highlights the missing fields.
- If invalid information is entered, the system asks the student to correct it.
- If a system error occurs, the complaint is not confirmed as successfully submitted.

Postcondition

A valid complaint record exists with status Submitted.

UC-006 - Assign Complaint

Primary Actor: Hostel/Warden Authority

Preconditions

- Authority is authenticated.
- Complaint exists.
- Complaint is eligible for assignment.

Main Flow

1. Authority opens the complaint management area.
2. Authority selects a complaint.
3. System displays complaint details.
4. Authority selects appropriate maintenance staff.
5. System verifies authorization.
6. System records the assignment.
7. System changes complaint status to Assigned.
8. System records the assignment in complaint history.
9. System makes the assignment available to the maintenance staff.

Alternative/Exception Flows

- If no suitable maintenance staff is available, the complaint remains unassigned.
- If the user is unauthorized, the assignment operation is rejected.
- If the complaint is already closed, assignment is rejected.

Postcondition

The complaint is assigned to the selected maintenance staff.

UC-008 - Resolve Complaint

Primary Actor: Maintenance Staff / Hostel Authority

Preconditions

- User is authenticated.
- Complaint is assigned/in progress.
- User has authorization to update the complaint.

Main Flow

1. Staff opens the assigned complaint.

2. System displays complaint details.
3. Staff performs or coordinates the required maintenance work.
4. Staff enters resolution details.
5. Staff marks the issue as resolved.
6. System validates the resolution information.
7. System changes status to Resolved.
8. System records the resolution.
9. System makes the updated status visible to the concerned student.

Alternative/Exception Flows

- If resolution information is missing, the system rejects the update.
- If the issue cannot yet be resolved, staff records progress and leaves the complaint in an active status.

Postcondition

The complaint has recorded resolution information and status Resolved.

20. User Stories

The requirements will also be represented in Agile form.

US-001

As a user, I want to log in so that I can securely access the system.

US-002

As a student, I want to submit a complaint so that my hostel issue can be reported.

US-003

As a student, I want to provide complaint details and category so that my issue can be understood correctly.

US-004

As a student, I want to view my complaint status so that I know its current progress.

US-005

As a student, I want to view my complaint history so that I can track my previous complaints and outcomes.

US-006

As a hostel authority, I want to review submitted complaints so that hostel issues can be assessed and managed.

US-007

As a hostel authority, I want to assign complaints to maintenance staff so that responsibility for the issue is clear.

US-008

As maintenance staff, I want to view complaints assigned to me so that I know which issues require action.

US-009
As maintenance staff, I want to update complaint progress so that the current work status is recorded.

US-010
As maintenance staff, I want to record resolution details so that completed work is documented.

US-011
As a user, I want to receive complaint status updates so that I am informed about important changes.

US-012
As an administrator, I want to manage users and roles so that access to the system is controlled.

US-013

As an authorized user, I want to generate complaint reports so that complaint activity can be monitored.

US-014

As a hostel authority, I want to close a resolved complaint so that completed complaints are formally recorded as closed.

21. Acceptance Criteria Example

US-004 -Submit Complaint

ID	Acceptance Criterion
AC-01	Students must be authenticated
AC-02	Required complaint fields must be provided.
AC-03	Complaint category must be valid.
AC-04	A unique complaint ID must be generated.
AC-05	Initial complaint status must be Submitted.
AC-06	Complaint must be stored successfully.
AC-07	Student must receive submission confirmation.

US-007 - Assign Complaint

ID	Acceptance Criterion
AC-08	Authority must be authenticated.
AC-09	Authority must have permission to assign complaints.
AC-10	Selected complaint must exist and be assignable.
AC-11	Selected maintenance staff must be valid/active.
AC-12	Assignment must be recorded.
AC-13	Complaint status must become Assigned.
AC-14	Assignment must be visible to the responsible staff.

US-009 -Update Complaint Progress

ID	Acceptance Criterion

AC-15	Maintenance staff must be authenticated.
AC-16	Staff must be authorized for the complaint.
AC-17	Progress information must be provided.
AC-18	Invalid required information must be rejected.
AC-19	Updated progress/status must be stored.
AC-20	Student/authority must be able to view the updated status.

22. Requirements Prioritization

Must Have

- Authentication;
- role-based access;
- complaint submission;
- complaint categorization;
- complaint review;
- complaint assignment;
- status tracking;
- maintenance progress updates;
- complaint resolution;
- complaint closure;
- complaint history.

Should Have

- Notifications/status updates;
- basic reports/dashboard;
- user/role administration improvements.

Could Have

- Complaint attachments;
- advanced filtering;
- richer analytics;
- escalation reminders.

Won't Have in Initial Release

- Payment processing;
- mobile application;
- AI-based complaint diagnosis/recommendation;
- IoT-based repair;
- complete hostel admission/room allocation;
- complete college ERP;
- academic course registration;
- online examinations.

23. Requirements Traceability

Requirements Traceability Matrix (RTM).

An initial version:

Requirement	User Story	Use Case	Test
FR-001	US-001	UC-001	TBD
FR-006	US-002	UC-003	TBD
FR-008	US-002	UC-004	TBD
FR-011	US-006	UC-005	TBD
FR-013	US-007	UC-006	TBD
FR-015	US-009	UC-007	TBD
FR-017	US-010	UC-008	TBD
FR-021	US-013	UC-012	TBD

24. Requirements Validation

Correctness

Requirements represent genuine needs identified in Assignment 1 for students, hostel authorities, maintenance staff and administrators.

Unambiguity

Requirements identify clear actors, actions and outcomes and avoid vague wording.

Completeness

Major in-scope functionality from Assignment 1 is represented through requirements, rules, use cases, stories or data requirements.

Consistency

Requirements are consistent with the defined user roles, complaint lifecycle and project scope.

Verifiability

Requirements are written so that future test cases can determine whether they are satisfied.

Feasibility

Requirements are appropriate for the student team's project constraints and planned prototype.

Traceability

Functional requirements are mapped to user stories and use cases, with future testing links planned.

Prioritization

MoSCoW prioritization identifies what is essential for the initial release.

Modifiability

Unique IDs and modular sections allow requirements to be updated systematically.

25. Example of Requirement Quality

Poor

- The system should be fast and user-friendly.
- This is ambiguous and difficult to verify.

Better

- The system shall provide consistent navigation across major user interfaces.
- The system should provide acceptable response times for normal user operations under expected workload

26. Requirement Change Management

Because the project uses an incremental development approach, requirements may change.
For example:
Change Request: Add photo/video evidence to complaints.
The team should follow:

Change Request
↓
Impact Analysis
↓
Requirement Update
↓
Backlog / Prioritization
↓
Design Change
↓
Implementation
↓
Testing
↓
Release

27. Future Requirements

Potential future requirements include:

- Mobile application.
- Advanced complaint analytics.
- Automated escalation for overdue complaints.
- Photo/video evidence for complaints.
- Integration with institutional authentication.
- Integration with institutional ERP.
- AI-assisted complaint categorization or routing.
- Automated maintenance reminders.

28. SRS Baseline and Versioning

- **Initial baseline:** SRS v1.0
- **Version Control Information:** SRS v1.0 defines the initial user classes, system context, functional and non-functional requirements, business rules, data requirements, complaint

lifecycle, use cases, user stories, acceptance criteria, prioritization and traceability for the Hostel Complaint Management System. It establishes the requirements baseline for subsequent UML-based analysis and design.

29. Individual Contribution

- DHRUV AGRAWAL - Non-functional requirements; data requirements; requirements validation.
- SANYAM HUMAD - System context; complaint lifecycle; detailed use cases.
- HARSHIL AGRAWAL - Functional requirements; business rules; requirements traceability.
- AKSHAT KHANDAL - User classes; user stories; acceptance criteria; MoSCoW prioritization.

We declare that:

• We have disclosed the use of AI tools, if any, during preparation of this assignment.
• AI-generated content, if used, has been reviewed, verified, edited and adapted by the team.
• We understand the project sufficiently to explain and defend all submitted content during a viva/discussion.
• We have not submitted AI-generated content blindly or without verification.
• We have not used AI to misrepresent work performed by the team.
• The final document reflects the team's own project understanding and decisions.

AI Tool(s) Used: ChatGPT

Purpose of AI Use Used for understanding the assignment guidelines, brainstorming and structuring project content, drafting initial sections, and reviewing/refining the assignment. All AI-assisted content was reviewed, verified, edited and adapted by the team.

Team ID: T042

Team Member Names: DHRUV AGRAWAL, SANYAM HUMAD, HARSHIL AGRAWAL, AKSHAT KHANDAL

Date: 07/09/26

Team Declaration:

We confirm that the information provided above is accurate and that we accept responsibility for the submitted work.

Team Representative Signature/Confirmation: Sanyam Humad

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Lab Assignment 3: Solution

1. Three Scenarios

Scenario S-01 - Submit Complaint (UC-003)
Actors: Student and System.
Starting condition: The student is registered and authenticated, and the complaint submission function is available.
Main sequence:
The student selects Submit Complaint. The system displays the complaint form. The student enters the title, description, category, hostel/block, and room/location details, where applicable. The student submits the form. The system validates the mandatory information, generates a unique Complaint ID, records the submission date and time, stores the complaint, sets its status to Submitted, and displays a submission confirmation.
Alternative/exception:
If any information is missing or invalid, the system highlights the relevant fields and asks the student to correct them. If a system error occurs, the system does not display a false success message, and the complaint is not treated as submitted.
Expected outcome:
A valid complaint record exists with a unique Complaint ID, submission date and time, and status Submitted.

Scenario S-02 - Assign Complaint (UC-006)
Actors: Hostel/Warden Authority, Maintenance Staff, and System.
Starting condition: The authority is authenticated, the complaint exists, and the complaint is eligible for assignment.
Main sequence:
The authority opens complaint management, selects a complaint, and views its details. The authority selects a suitable maintenance staff member. The system verifies the authority's authorization and the staff member's validity, records the assignment and assignment time, changes the complaint status to Assigned, records the status change in complaint history, and makes the assignment visible to the selected maintenance staff member.
Alternative/exception:
Unauthorized users are rejected. If no suitable maintenance staff member is available, the complaint remains unassigned. A closed complaint cannot be assigned.
Expected outcome:
The complaint is assigned to a valid maintenance staff member, and its status becomes Assigned.

Scenario S-03 - Resolve Complaint (UC-008)
Actors: Maintenance Staff, optionally authorized Hostel/Warden Authority, and System.
Starting condition: The user is authenticated and authorized, and the complaint is in the **Assigned** or **In Progress** state.

Main sequence:
The staff member opens the assigned complaint. The system displays the complaint details. The staff member performs or coordinates the maintenance work and enters the progress and resolution details. The staff member marks the issue as resolved. The system validates the information, records the resolution details and resolution time, changes the complaint status to **Resolved**, records the status change in complaint history, and makes the updated outcome visible to the concerned student and authority.
Alternative/exception:
If the resolution information is missing or invalid, the system rejects the submission and asks the staff member to correct it. If the issue is not resolved, the system records the progress update and keeps the complaint status as In Progress. Unauthorized users cannot resolve the complaint.
Expected outcome:
The resolution information is stored, and the complaint status becomes Resolved. Closure of the complaint happens separately.

2. UML Use Case Diagram

Diagram Title :
UML Use Case Diagram - Hostel Complaint Management System
System Boundary
The diagram contains one large rectangle representing the Hostel Complaint Management System. All use cases are placed inside the rectangle, while all actors are placed outside the rectangle.
Actors
The following actors are included:

1. Student
2. Hostel/Warden Authority
3. Maintenance Staff
4. Administrator
5. Institute Management

Student Use Cases
The Student actor is connected to:

- Submit Complaint (UC-003)
- View Complaint Status
- View Complaint History
- View Notification
- Logout

Hostel/Warden Authority Use Cases
The Hostel/Warden Authority actor is connected to:

- Review Complaint
- Categorize/Prioritize Complaint
- Assign Complaint (UC-006)
- Close Complaint
- Generate Complaint Reports
- Logout

Maintenance Staff Use Cases

The Maintenance Staff actor is connected to:

- View Assigned Complaints
- Update Complaint Progress
- Resolve Complaint (UC-008)
- Logout

Administrator Use Cases

The Administrator actor is connected to:

- Manage Users and Roles
- Generate Complaint Reports
- Logout

Institute Management Use Cases

The Institute Management actor is connected to:

- Generate Complaint Reports

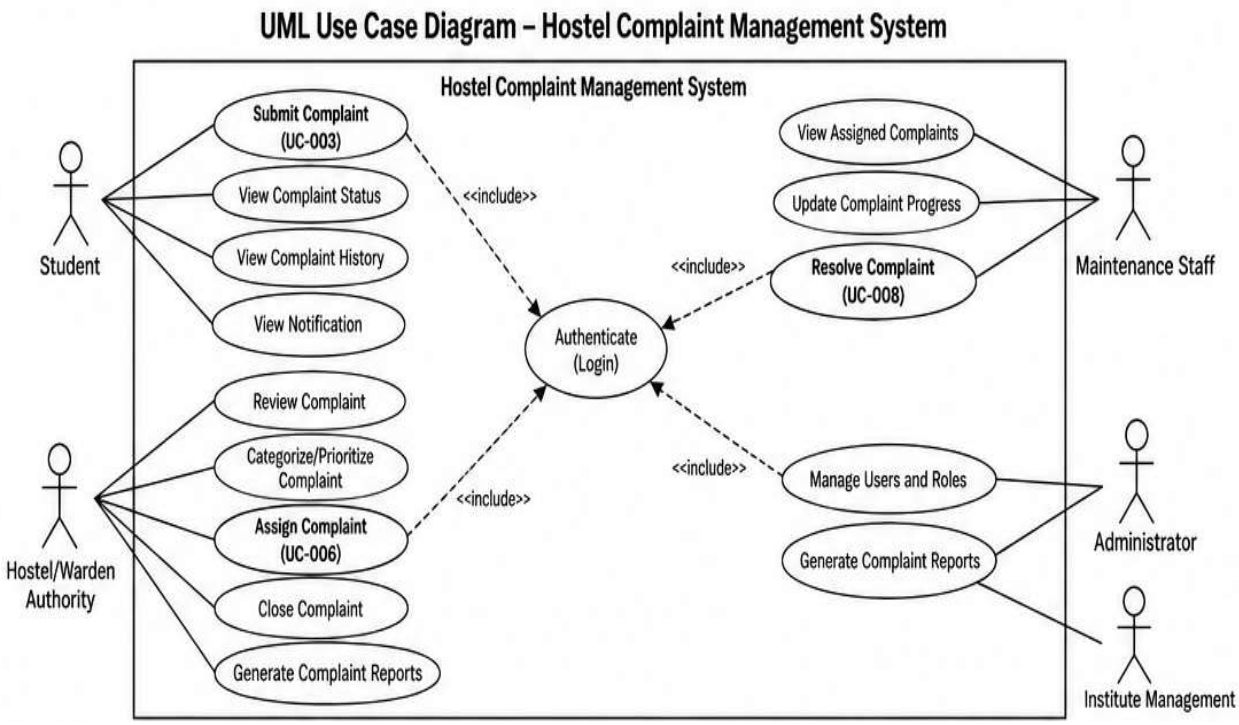
Authentication

The central use case is:

Authenticate (Login)

The following use cases may include authentication:

- Submit Complaint
- Assign Complaint
- Resolve Complaint
- Manage Users and Roles



3. UML Activity Diagram - UC-003 Submit Complaint

Diagram Title

Activity Diagram - UC-003 Submit Complaint

Swim-lanes

The activity diagram contains two swim-lanes:

1. Student
2. System

Workflow Description

The activity starts with an initial node in the Student lane.

Student Lane

1. Student Login
2. Enter Complaint Details
3. Submit Complaint

The complaint details include:

- Category
- Title
- Description
- Hostel/Block
- Room/Location

System Lane

1. Authenticate Student
2. Validate Mandatory Information
3. Generate Unique Complaint ID
4. Record Submission Date/Time
5. Create Complaint Record
6. Set Status to Submitted
7. Display Submission Confirmation

Decision

The system checks:

All Required Information Complete?

No Branch

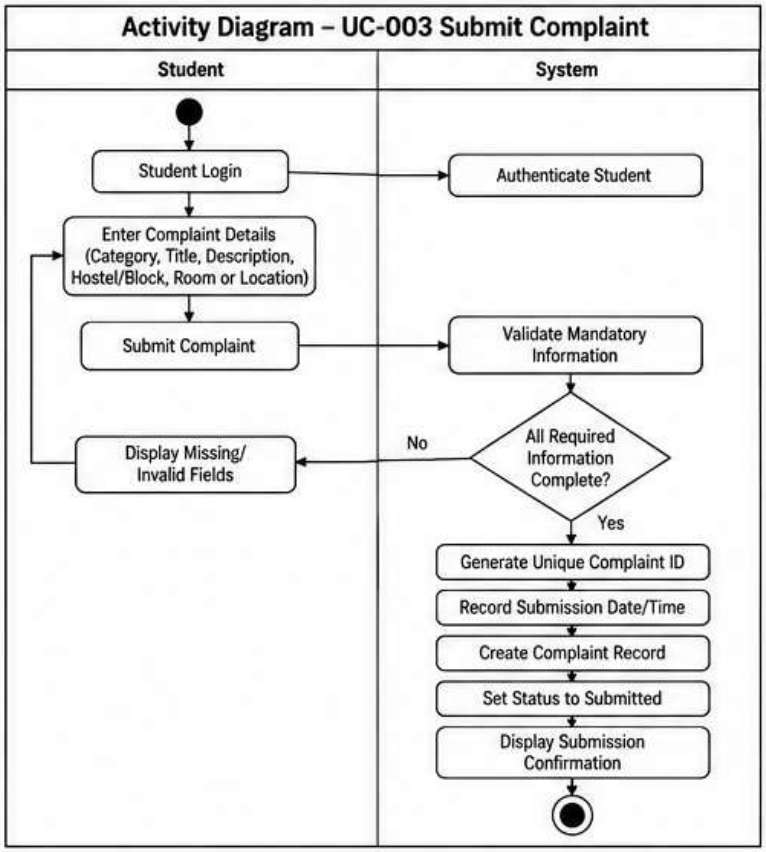
If the information is incomplete:

1. Display Missing/Invalid Fields.
2. Return to Enter Complaint Details.

Yes Branch

If all information is valid:

1. Generate Unique Complaint ID.
2. Record Submission Date/Time.
3. Create Complaint Record.
4. Set Status to Submitted.
5. Display Submission Confirmation.
6. End the activity.



4. UML Activity Diagram - UC-006 Assign Complaint

Diagram Title

Activity Diagram - UC-006 Assign Complaint

Swim-lanes

The activity diagram contains three swim-lanes:

1. Hostel/Warden Authority
2. System
3. Maintenance Staff

Workflow Description

The activity starts in the Hostel/Warden Authority lane.

Hostel/Warden Authority Lane

1. Authority Login
2. Open Complaint Management
3. Select Complaint
4. Select Maintenance Staff

System Lane

1. Authenticate Authority
2. Display Complaint Details
3. Check Complaint Eligibility
4. Verify Staff Selection and Authorization
5. Record Complaint Assignment
6. Set Status to Assigned

7. Record Assignment in History

Maintenance Staff Lane

1. View Assigned Complaint

Decisions

Decision 1: Complaint Assignable?

The system checks whether the complaint is eligible for assignment.

Yes

The authority selects a maintenance staff member.

No

The system displays:

Complaint Cannot Be Assigned

The activity then ends without assigning the complaint.

Decision 2: Valid Staff Selection?

The system checks whether the selected staff member is valid and authorized.

Yes

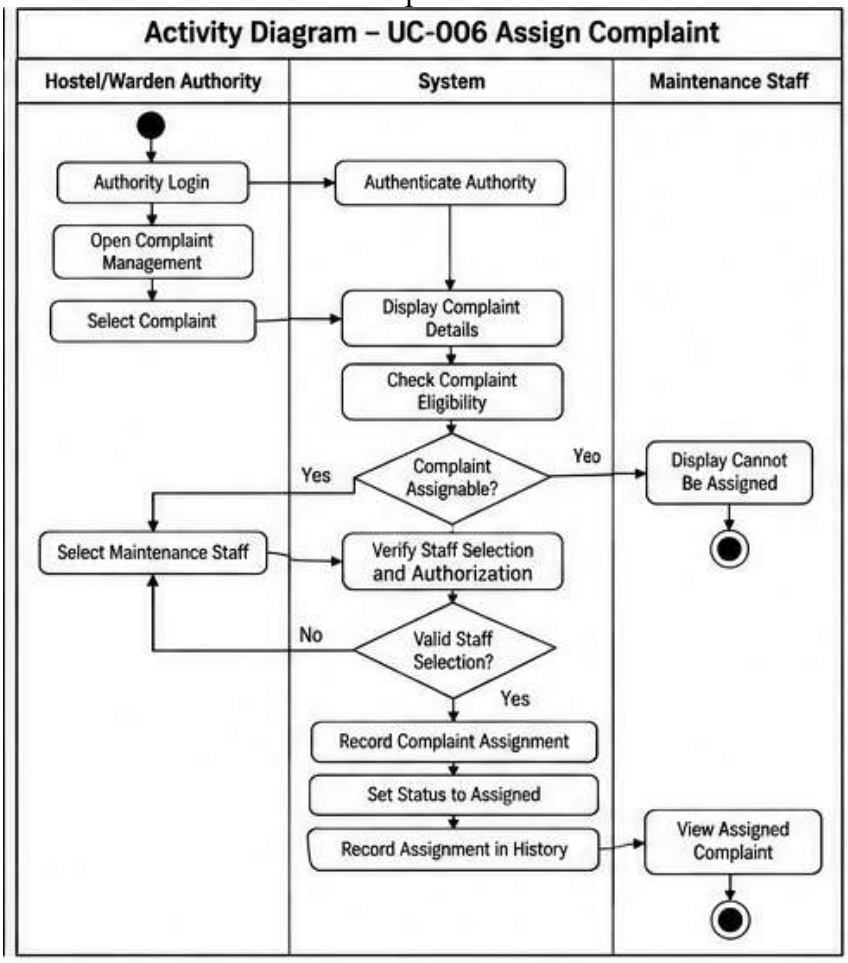
The system records the assignment and changes the status to **Assigned**.

No

The system displays:

Display Selection Error

The authority is returned to the staff selection step.



5. UML Activity Diagram - UC-008 Resolve Complaint

Diagram Title

Activity Diagram - UC-008 Resolve Complaint

Swim-lanes

The activity diagram contains three swim-lanes:

1. Maintenance Staff
2. System
3. Student/Hostel/Warden Authority

Workflow Description

The activity starts in the Maintenance Staff lane.

Maintenance Staff Lane

1. Maintenance Staff Login
2. Open Assigned Complaint
3. Perform Maintenance Work
4. Enter Progress and Resolution Details
5. Mark Issue as Resolved

System Lane

1. Authenticate Maintenance Staff
2. Display Complaint Details
3. Verify Assignment and Authorization
4. Record Progress Update
5. Validate Resolution Details
6. Record Resolution Details
7. Set Status to Resolved
8. Record Resolution in History

Student/Hostel/Warden Authority Lane

1. View Updated Complaint Status

Decision 1: Authorized User?

The system verifies whether the user is authorized to resolve the complaint.

No

The system displays:
Reject Operation
The activity ends.

Yes

The maintenance staff member is allowed to continue.

Decision 2: Issue Resolved?

The system checks whether the issue has been resolved.

No

1. Record the progress update.
2. Keep the complaint status as **In Progress**.
3. Return to the maintenance work step.

Yes

1. Enter resolution details.
2. Validate the resolution information.
3. Record resolution details.

4. Set the status to **Resolved**.
5. Record the resolution in complaint history.
6. Display the updated status to the student and authorized authority.
7. End the activity.

Decision 3: Resolution Information Complete?

No

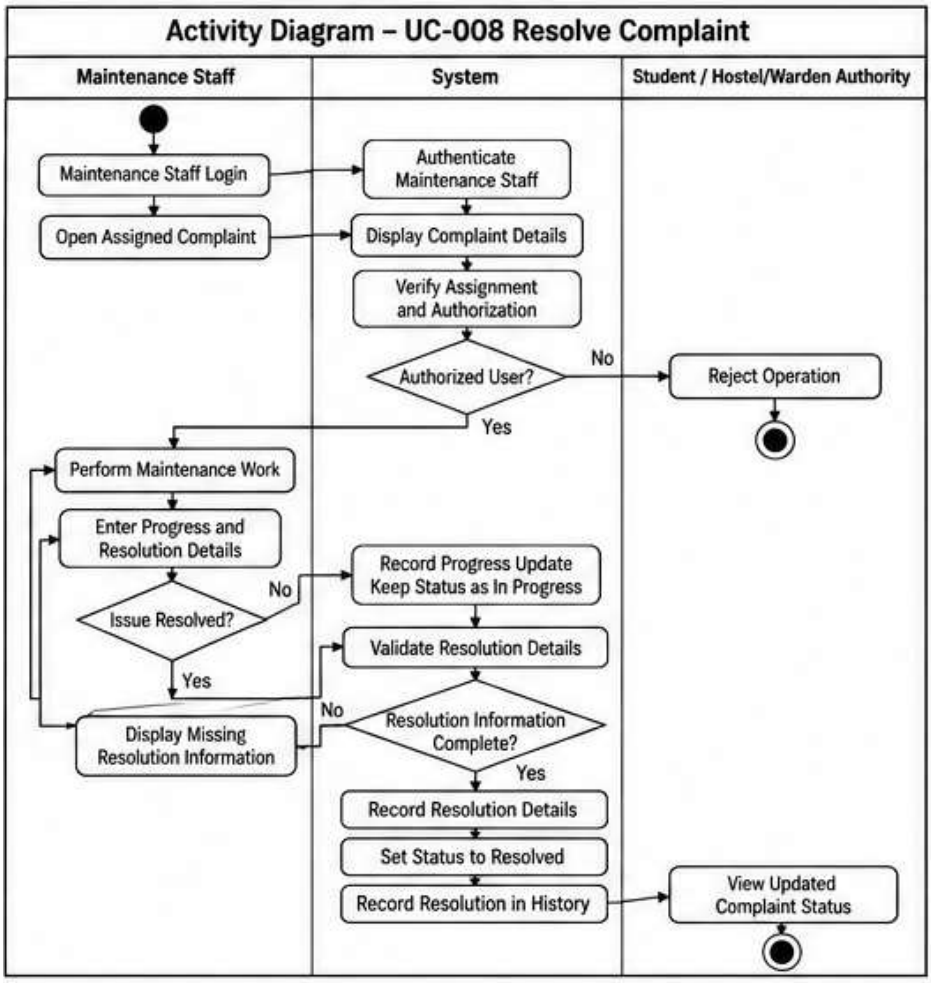
The system displays:

Display Missing Resolution Information

The staff member is returned to the resolution information step.

Yes

The system records the resolution and changes the status to **Resolved**.



6. Individual Contribution

- **DHRUV AGRAWAL [24ucs016]:** Review of system requirements, activity-flow validation and consistency checking.
- **SANYAM HUMAD [24ucs017]:** Preparation of scenarios, system-level UML modelling and activity diagram coordination.
- **HARSHIL AGRAWAL [24ucs018]:** Use-case identification, actor associations and requirements-to-use-case consistency.
- **AKSHAT KHANDAL [24ucs019]:** Activity diagram preparation, decision-flow checking and documentation formatting.

7. AI Usage Declaration

AI Tool(s) Used: ChatGPT

Purpose of AI Use:

AI tools were used for understanding the assignment instructions, organizing the scenarios, preparing initial textual UML descriptions and reviewing the structure of the diagrams.

All AI-assisted content was reviewed, verified, edited and adapted by the team. The team is responsible for the correctness and understanding of the submitted scenarios and UML models.

We declare that:

1. We have disclosed the use of AI tools during preparation of this assignment.
2. AI-generated suggestions were reviewed, verified and adapted by the team.
3. We understand and can explain the submitted scenarios and UML diagrams.
4. We verified that the UML models are consistent with the project's requirements, use cases and scope.
5. The final submission represents the team's own understanding and modelling decisions.
6. We have not used AI to misrepresent work performed by the team.

Team ID: T042

Team Member Names: DHRUV AGRAWAL, SANYAM HUMAD, HARSHIL AGRAWAL, AKSHAT KHANDAL

Date: 13/09/26

Team Declaration:

We confirm that the information provided above is accurate and that we accept responsibility for the submitted work.

Team Representative Signature/Confirmation: Sanyam Humad

The LNM Institute of Information Technology, Jaipur
Software Development Laboratory
Course Code: CS314
Lab Assignment 4: Solution

1. Classes

The class model is derived from same three detailed use cases selected in Assignments 2 and 3:

- **UC-003 - Submit Complaint**
- **UC-006 - Assign Complaint**
- **UC-008 - Resolve Complaint**

The classes represent the main complaint-management entities and the responsibilities required to submit complaints, categorize them, assign them to maintenance staff, track status changes, record resolutions and notify concerned users.

The model contains **10 meaningful classes**. The classes are derived from the user classes, functional requirements, data requirements and detailed use cases established in the previous assignments.

Class Name	Important Attributes	Operations / Responsibilities	Traceability
User	userId, name, email, contactInformation, role, accountStatus	login(), logout(), updateAccountInformation()	UC-003, UC-006, UC-008
Student	studentId	submitComplaint(), viewStatus(), viewHistory(), viewResolution()	UC-003
HostelWardenAuthority	authorityId, role	reviewComplaint(), categorizeComplaint(), prioritizeComplaint(), assignComplaint(), monitorComplaint(), closeComplaint()	UC-006
MaintenanceStaff	staffId, specialization, accountStatus	viewAssignedComplaints(), updateProgress(), recordResolution(), markCompleted()	UC-008
Complaint	complaintId, title, description, hostelBlock, roomLocation, priority, submissionDateTime, currentStatus	validateComplaint(), create(), generateComplaintId(), updateStatus(), retrieveHistory()	UC-003, UC-006, UC-008

ComplaintCategory	categoryId, categoryName, description, activeStatus	validateCategory(), classifyComplaint(), checkAvailability()	UC-003, UC-006
ComplaintAssignment	assignmentId, complaintId, staffId, assignedBy, assignmentDateTime	assignStaff(), validateAssignment(), recordAssignment()	UC-006
ComplaintStatusHistory	historyId, complaintId, previousStatus, newStatus, changedBy, changeDateTime, comments	recordStatusChange(), retrieveHistory(), displayTimeline()	UC-006, UC-008
Resolution	resolutionId, complaintId, resolutionDetails, resolvedBy, resolutionDateTime	validateResolution(), recordResolution(), updateResolutionStatus()	UC-008
Notification	notificationId, recipientId, complaintId, message, notificationType, readStatus, dateTime	sendNotification(), markAsRead(), displayNotification()	UC-003, UC-006, UC-008

2. Object Diagram

2.1) Object Diagram Scenario

The Object Diagram represents a snapshot of **UC-006 - Assign Complaint** after a student has submitted a complaint and the Hostel/Warden Authority is assigning it to suitable maintenance staff.

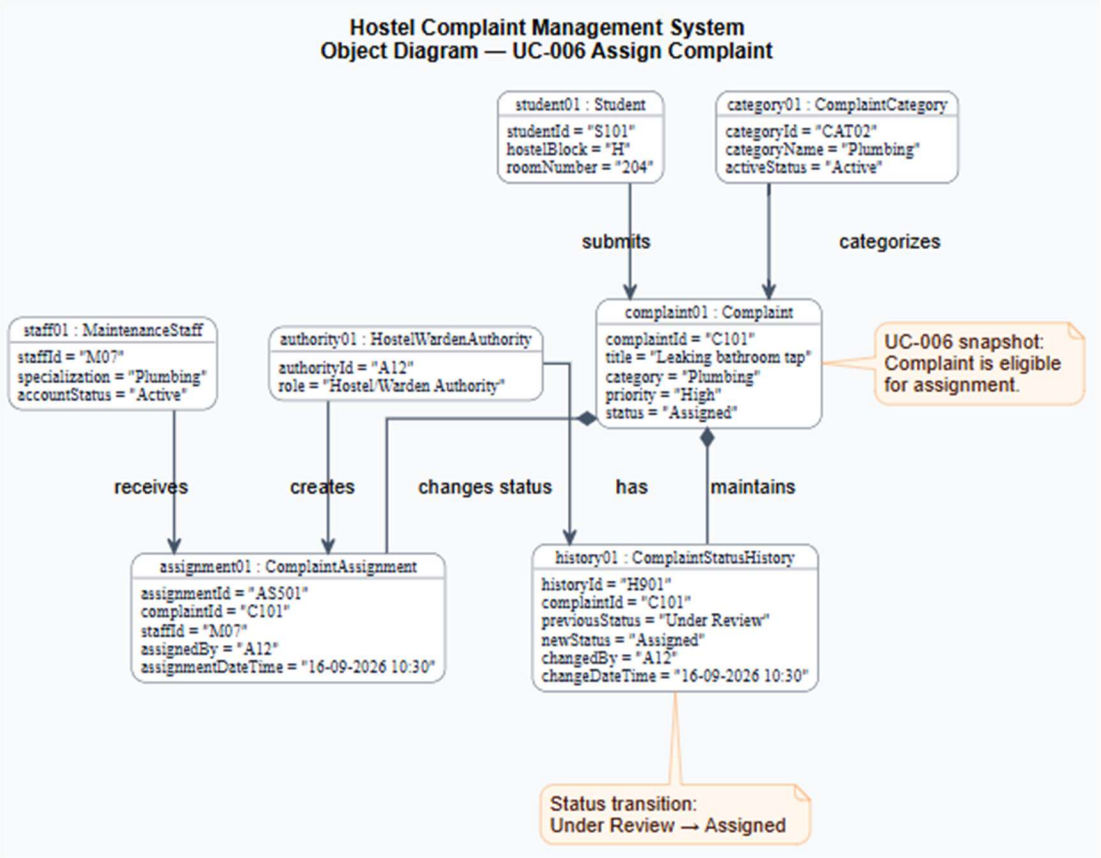
In the snapshot, student S101 has submitted complaint C101 regarding a leaking bathroom tap. The complaint belongs to the Plumbing category and is currently eligible for assignment.

The Hostel/Warden Authority A12 selects maintenance staff member M07, who is responsible for plumbing-related maintenance. The system creates assignment AS501, changes the complaint status from Under Review to Assigned, and records the status change in complaint history.

This follows the existing UC-006 workflow: the authority selects a complaint, selects suitable maintenance staff, the system verifies authorization, records the assignment, changes the status to Assigned and records the change in complaint history.

Representative objects
student01 : Student
complaint01 : Complaint
category01 : ComplaintCategory
authority01 : HostelWardenAuthority
staff01 : MaintenanceStaff
assignment01 : ComplaintAssignment
history01 : ComplaintStatusHistory

2.2 Object Diagram



3. Relationships

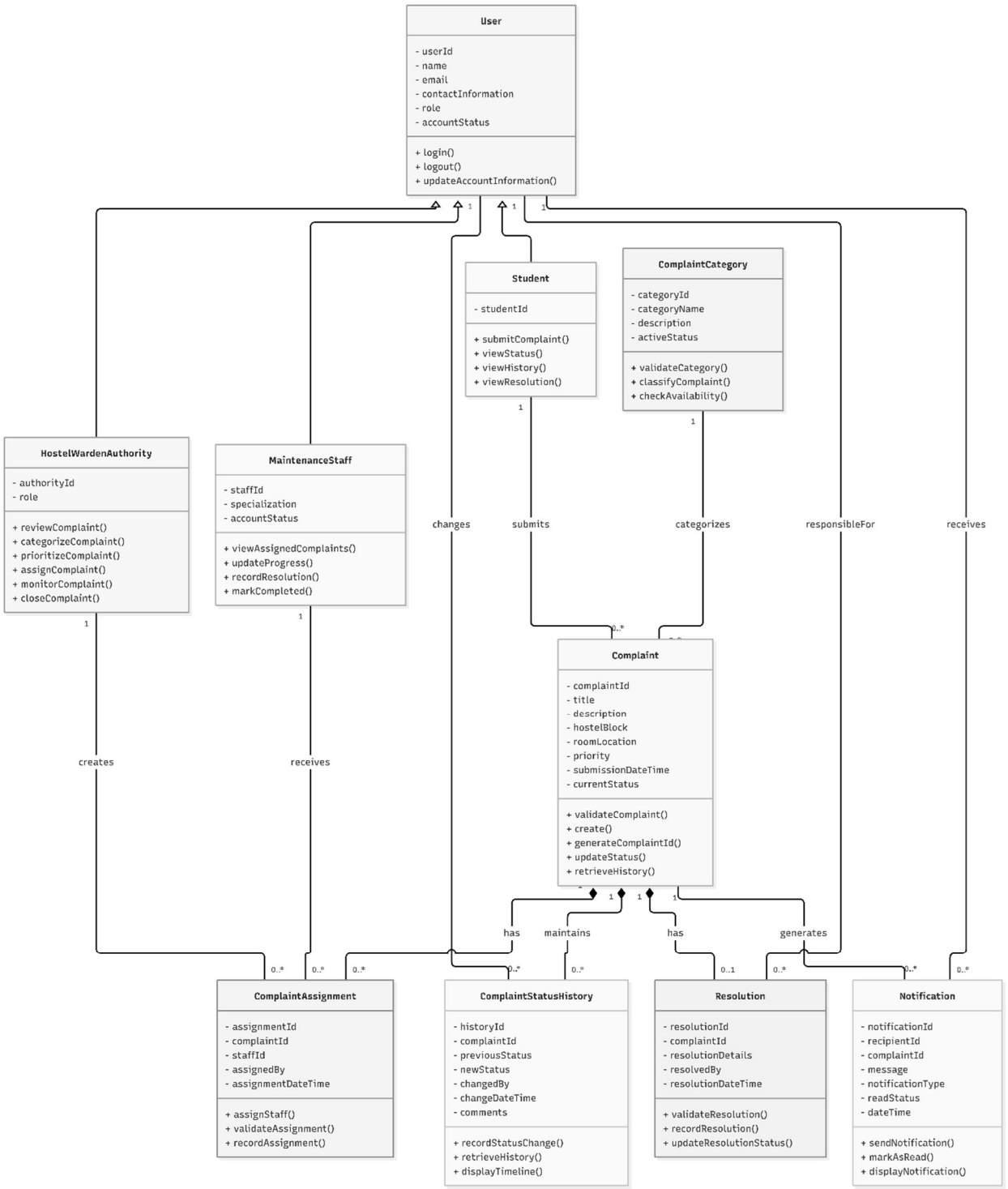
The relationships are derived from the requirements, data model and the three selected use cases.
Association is used when two classes need to reference or interact with each other during normal system operation.
Composition is used where a dependent record belongs to a particular complaint and does not represent an independent complaint-management record in the modeled workflow.
The selected workflows establish submission, assignment, status tracking and resolution relationships.

Classes	Relationship Type	Multiplicity	Justification
User - Student	Generalization		Student is a specialized system user.
User -Hostel/Warden Authority	Generalization		Hostel/Warden Authority is a specialized system user.
User -Maintenance Staff	Generalization		Maintenance Staff is a specialized system user.
Student -Complaint	Association	1 Student : 0..* Complaints	A student may submit multiple complaints; each complaint belongs to one student.
Complaint Category - Complaint	Association	1 Category : 0..* Complaints	One category can classify multiple complaints.
Complaint -Complaint Assignment	Composition	1 Complaint : 0..* Assignments	Assignment records belong to the complaint.
Maintenance Staff - Complaint Assignment	Association	1 Staff : 0..* Assignments	A maintenance staff member may receive multiple assignments.
Hostel/Warden Authority -Complaint Assignment	Association	1 Authority : 0..* Assignments	An authorized authority may create multiple assignments.
Complaint -Complaint Status History	Composition	1 Complaint : 0..* History Records	Status changes are maintained as part of the complaint history.
User -Complaint Status History	Association	1 User : 0..* History Records	Changed By identifies the user who performed the status change.
Complaint - Resolution	Composition	1 Complaint : 0..1 Resolution	A complaint may have no resolution initially and one recorded resolution after resolution.
User -Resolution	Association	1 User : 0..* Resolutions	A user can be responsible for multiple resolution records.
Complaint - Notification	Association	1 Complaint : 0..* Notifications	A complaint may generate multiple notifications.
User -Notification	Association	1 User : 0..* Notifications	A user may receive multiple notifications.

4. UML Class Diagram

The following UML Class Diagram represents the identified classes and their relationships. It uses standard UML class notation with attributes, operations, associations, inheritance and multiplicities.

Class Diagram:



5. Individual Contribution

• Dhruv Agrawal: class identification and class responsibilities.
• Sanyam Humad: object snapshot and consistency with UC-006.
• Harshil Agrawal: relationships, multiplicities and UML semantics.
• Akshat Khandal: PlantUML formatting, readability and documentation checking

6. **AI Usage Declaration**

AI tools may be used as supporting tools for this assignment, but the submitted work must represent the team's own understanding and contribution.
We declare that:

- We have disclosed the use of AI tools during preparation of this assignment.
- AI-assisted content, if used, has been reviewed, verified, edited and adapted by the team.
- We understand the project sufficiently to explain and defend all submitted classes, objects and relationships during a viva/discussion.
- We have verified that the UML models are consistent with the project's requirements, use cases and scope.
- We have not submitted AI-generated content blindly or without verification.
- We have not used AI to misrepresent work performed by the team.
- The final submission reflects the team's own project understanding, contribution and modelling decisions.

AI Tool(s) Used: ChatGPT/Mermaid AI
Purpose of AI Use: Understanding the assignment instructions, structuring the UML class/object model, reviewing traceability and preparing initial PlantUML representations. All AI-assisted content was reviewed, verified and adapted by the team.
Team ID: T042
Team Members:
DHRUV AGRAWAL, SANYAM HUMAD, HARSHIL AGRAWAL, AKSHAT KHANDAL
Date: 20/09/2026
Team Declaration:
We confirm that the information provided above is accurate and that we accept responsibility for the submitted work.

The LNM Institute of Information Technology, Jaipur
Software Development Laboratory
Course Code: CS314
Lab Assignment 5: Solution

1. Collaboration Diagrams

1.1 UC-003 – Submit Complaint

Purpose:
Models the interaction between the Student and the complaint-management objects when a new hostel complaint is submitted.

Participating Objects:
Student, Complaint, ComplaintCategory, ComplaintStatusHistory, and Notification.

Interaction:
The student submits the complaint. The system validates the complaint details and category, generates a unique complaint ID, creates the complaint record, stores the initial status as *Submitted*, and sends a confirmation notification to the student.

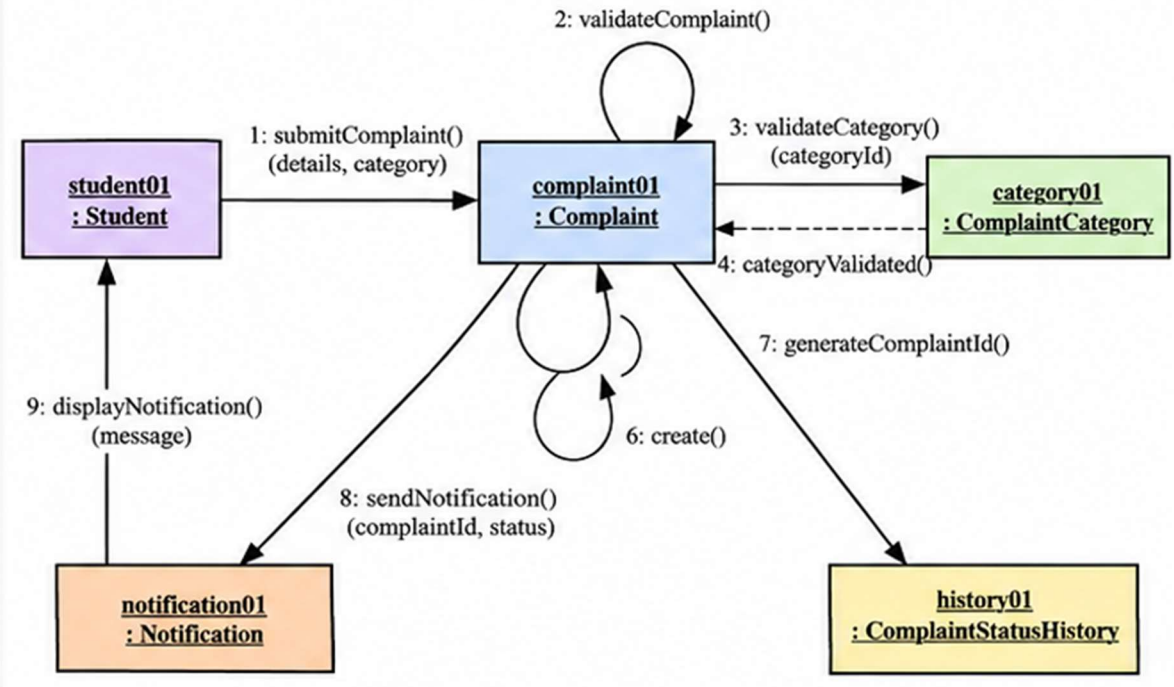


Figure 5.1: Collaboration Diagram – UC-003 Submit Complaint

Key Messages:

1. submitComplaint()
2. validateComplaint()
3. validateCategory()
4. generateComplaintId()
5. create()

Outcome:
A valid complaint is created with the initial status *Submitted*, and the student receives a confirmation notification.

1.2 UC-006 – Assign Complaint

Purpose:
Models how the Hostel/Warden Authority assigns an eligible complaint to maintenance staff.
Participating Objects:
HostelWardenAuthority, Complaint, MaintenanceStaff, ComplaintAssignment, ComplaintStatusHistory, and Notification.
Interaction:
The authority reviews the complaint, selects a suitable maintenance staff member, validates the assignment, changes the complaint status to *Assigned*, records the status change in the complaint history, and notifies the selected staff member.

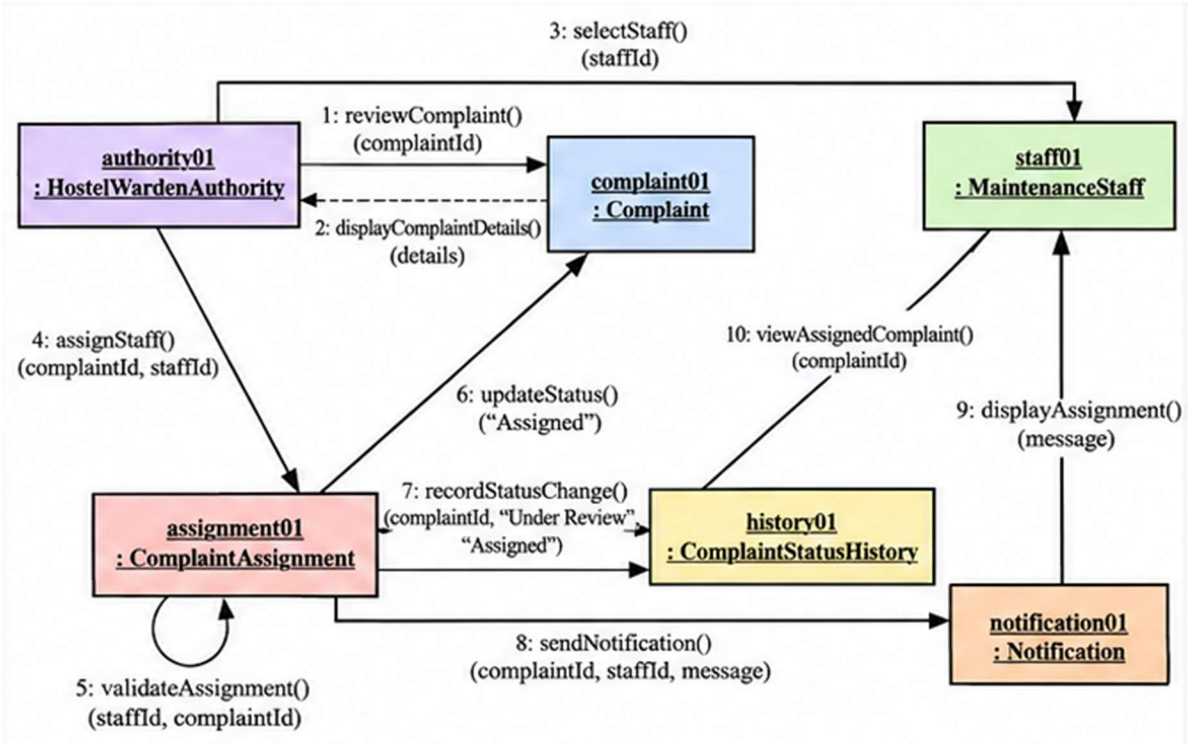


Figure 5.2: Collaboration Diagram – UC-006 Assign Complaint

Key Messages:

1. reviewComplaint()
2. selectStaff()
3. assignStaff()
4. validateAssignment()
5. updateStatus(Assigned)

Outcome:
A valid ComplaintAssignment is created, and the assigned maintenance staff member is notified.

1.3 UC-008 – Resolve Complaint

Purpose:
Models how Maintenance Staff handles an assigned complaint and records its resolution.
Participating Objects:
MaintenanceStaff, Complaint, Resolution, ComplaintStatusHistory, Notification, and Student.
Interaction:
The maintenance staff views the assigned complaint, updates its progress, records and validates

the resolution details, changes the complaint status to *Resolved*, updates the complaint history, and notifies the student.

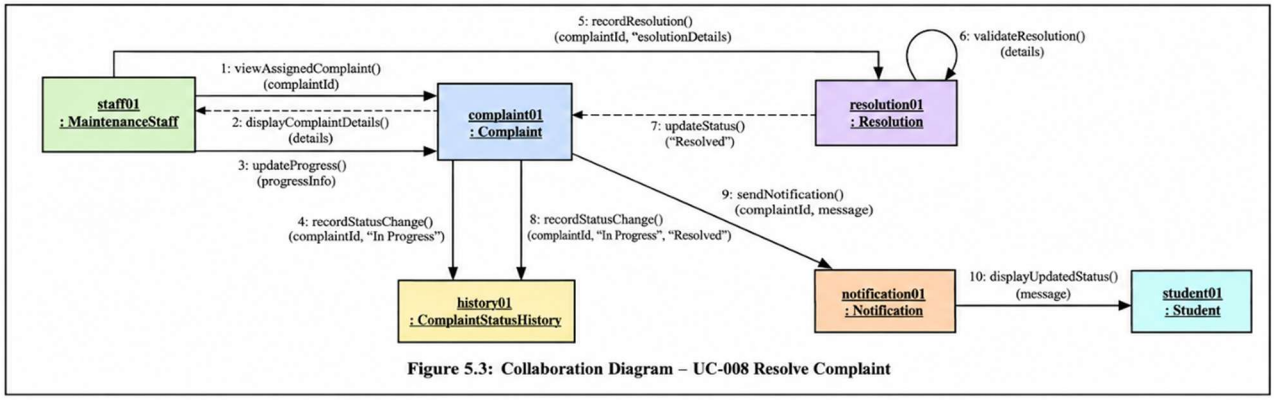


Figure 5.3: Collaboration Diagram – UC-008 Resolve Complaint

Key Messages:

1. viewAssignedComplaint()
2. updateProgress()
3. recordResolution()
4. validateResolution()
5. updateStatus(Resolved)

Outcome:

The resolution is recorded, and the student receives an update regarding the complaint status.

2. State Diagrams

2.1 Complaint State Diagram

Purpose:

Represents the lifecycle of the Complaint object from successful submission to final closure.

States:

Submitted, Under Review, Assigned, In Progress, Resolved, and Closed.

Transition Logic:

Review and assignment require the appropriate authority and a valid staff selection. Resolution requires valid resolution information, while closure is performed by the authorized authority.

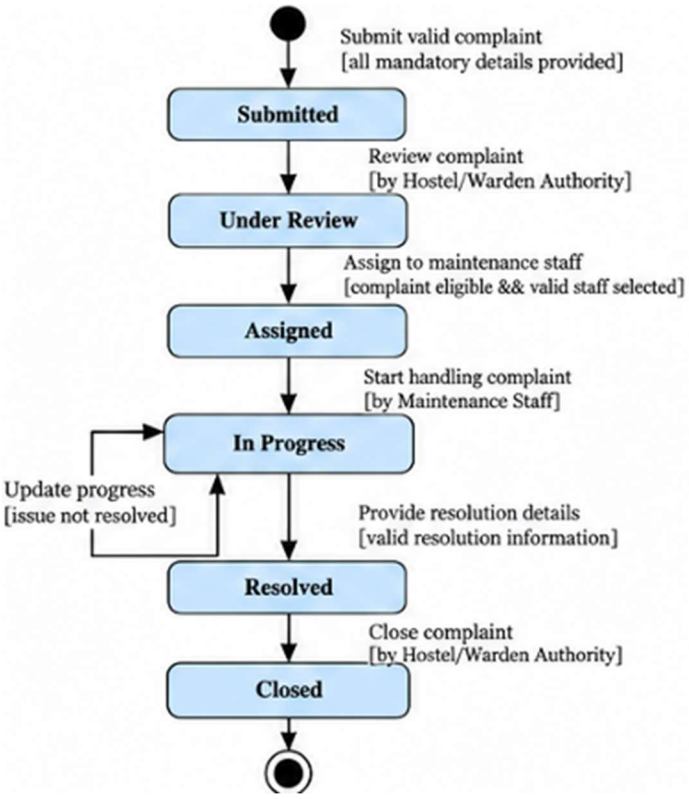


Figure 5.4: State Diagram – Complaint

Important Transition:

The complaint may remain in the *In Progress* state while work continues or when the issue has not yet been resolved. Once valid resolution information is provided, the complaint transitions to *Resolved*.

Normal Lifecycle:

Submitted → Under Review → Assigned → In Progress → Resolved → Closed

2.2 ComplaintAssignment State Diagram

Purpose:

Represents the lifecycle of the assignment connecting a complaint with maintenance staff.

States:

Unassigned, Assignment Pending, Assigned, In Progress, Completed, and Cancelled.

Transition Logic:

A valid staff selection creates the assignment. Staff acceptance initiates the work, and completion ends the normal assignment lifecycle. Invalid staff selection or authorized cancellation follows the corresponding alternative transition.

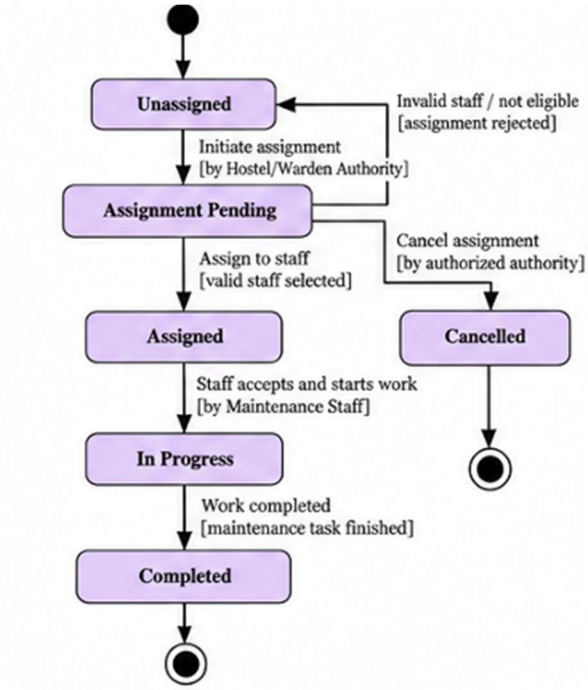


Figure 5.5: State Diagram – ComplaintAssignment

Normal Lifecycle:

Unassigned → Assignment Pending → Assigned → In Progress → Completed

Alternative Paths:

- The assignment may return to *Unassigned* when the selected staff member is invalid or ineligible.
- The assignment may transition to *Cancelled* when an authorized cancellation occurs.

3. UML Component Diagram

Purpose:

Shows the major software components supporting the three selected use cases and their dependencies.

Main Components:

1. Student Web Interface
2. Hostel Authority Interface
3. Maintenance Staff Interface
4. Authentication & Authorization Service
5. Complaint Management Service
6. Assignment Management Service
7. Resolution Management Service
8. Reporting Service
9. Database
10. Notification Service

Architecture:

The user interfaces access protected services through authentication and authorization. The management services persist complaint-related data in the database and use the Notification Service to communicate important complaint events.

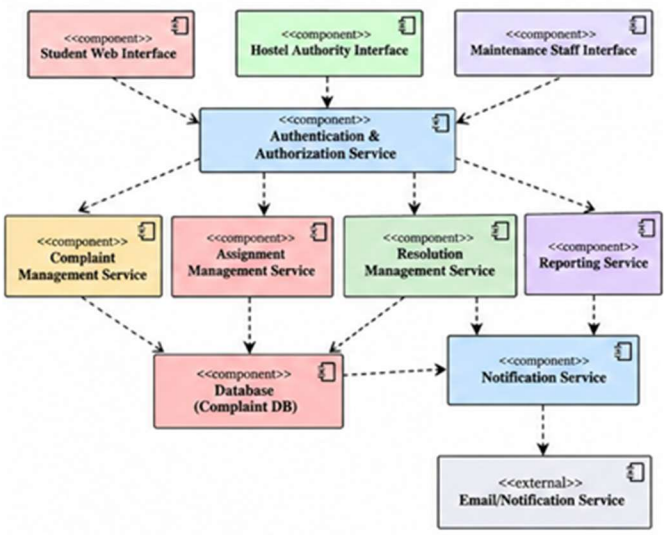


Figure 5.6: Component Diagram – Hostel Complaint Management System

Use-Case Support:

- **Complaint Management Service:** Supports complaint submission.
- **Assignment Management Service:** Supports the allocation of complaints to maintenance staff.
- **Resolution Management Service:** Supports complaint progress updates and resolution.
- **Reporting Service:** Provides authorized summaries and reports.
- **Notification Service:** Communicates important complaint updates and status changes to relevant users.

Dependency Flow:

User Interfaces → Authentication & Authorization Service → Management Services → Database

The management services also communicate with the Notification Service, which may use an external delivery service to send notifications.

4. Individual Contribution

- **Dhruv Agrawal:** Collaboration diagram modelling, interaction identification, and validation of object communication for the selected use cases.
- **Sanyam Humad:** State diagram modelling, state transitions, and consistency checking with the complaint workflow.
- **Harshil Agrawal:** Component identification, component dependencies, and UML relationship and interaction validation.
- **Akshat Khandal:** PlantUML formatting, diagram readability, documentation, and final consistency checking of all Assignment 5 models.

5. AI Usage Declaration

AI tools may be used as supporting tools for this assignment, but the submitted work must represent the team's own understanding and contribution.

We declare that:

- We have disclosed the use of AI tools during preparation of this assignment.
- AI-assisted content, if used, has been reviewed, verified, edited, and adapted by the team.

- We understand the project sufficiently to explain and defend all submitted collaboration diagrams, state diagrams, and component diagrams during a viva or discussion.
- We have verified that the UML models are consistent with the project's requirements, use cases, classes, objects, and project scope.
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- The final submission reflects the team's own project understanding, contribution, and modelling decisions.

AI Tool(s) Used: ChatGPT / Mermaid AI
Purpose of AI Use: Understanding the Assignment 5 instructions, structuring the collaboration, state, and component models, reviewing traceability, and preparing initial UML diagram representations. All AI-assisted content was reviewed, verified, and adapted by the team.
Team ID: T042
Team Members: Dhruv Agrawal, Sanyam Humad, Harshil Agrawal, Akshat Khandal
Date: 29/09/2026
Team Declaration: We confirm that the information provided above is accurate and that we accept responsibility for the submitted work.